NODE JS LAB

SIMPLE PROGRAM

1.Build a responsive web application for shopping cart with registration, login, catalog and cart pages using CSS3 features, flex and grid.

2. Make the above web application responsive web application using Bootstrap framework.

3. Use JavaScript for doing client – side validation of the pages implemented in experiment 1 and experiment 2.

4. Explore the features of ES6 like arrow functions, callbacks, promises, async/await. Implement an application for reading the weather information from openweathermap.org and display the information in the form of a graph on the web page
5. Create an xml for the bookstore. Validate the same using both DTD and XSD.

COMPLEX PROGRAM

6.Maintaining the transactional history of any user is very important. Explore the various session tracking mechanism (Cookies, HTTP Session)

7. Create a custom server using http module and explore the other modules of Node JS like OS, path, event.

8. Develop an express web application that can interact with REST API to perform CRUD operations on student data. (Use Postman)

9. For the above application create authorized end points using JWT (JSON Web Token).

10. Create a TODO application in react with necessary components and deploy it into GitHub.

Week 1
cart.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-
width, initial-scale=1.0">
<title>Cart - FBS</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header><h1>FBS E-Books</h1><p>World's Best
Online Books Website</p></header>
<nav>
  <a href="home.html">Home</a> <a
href="registration.html">Registration</a> <a
href="login.html">Login</a> <a
href="catalog.html">Catalog</a> <a
href="cart.html">Cart</a>
</nav>
<main>
  <h2 class="center">Shopping Cart</h2>
  <table>
    <tr><th>Book
Name</th><th>Price</th><th>Quantity</th><th>
Amount</th></tr>
    <tr><td>Java
Programming</td><td>₹500</td><td>1</td><td>₹
500</td></tr>
    <tr><td>Python
Programming</td><td>₹400</td><td>2</td><td>₹
800</td></tr>
    <tr><th colspan="3">Total
Amount</th><th>₹1300</th></tr>
  </table>
  <p class="center"><a class="button"
href="catalog.html">Continue Shopping</a></p>
</main>
<footer><p>© 2026 FBS E-Books. All Rights
Reserved.</p></footer>
</body>
</html>
```

catalog.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-
width, initial-scale=1.0">
<title>Catalog - FBS</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header><h1>FBS E-Books</h1><p>World's Best
Online Books Website</p></header>
<nav>
  <a href="home.html">Home</a> <a
href="registration.html">Registration</a> <a
href="login.html">Login</a> <a
href="catalog.html">Catalog</a> <a
href="cart.html">Cart</a>
</nav>
<main>
  <h2 class="center">Book Catalog</h2>
  <section class="book-grid">
    <div class="box"><h3>Java
Programming</h3><p>Price: ₹500</p><a
class="button" href="cart.html">Add to
Cart</a></div>
    <div class="box"><h3>Python
Programming</h3><p>Price: ₹400</p><a
class="button" href="cart.html">Add to
Cart</a></div>
    <div class="box"><h3>Web
Technologies</h3><p>Price: ₹300</p><a
class="button" href="cart.html">Add to
Cart</a></div>
  </section>
</main>
<footer><p>© 2026 FBS E-Books. All Rights
Reserved.</p></footer>
</body>
</html>
```

Home.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-
width, initial-scale=1.0">
<title>FBS E-Books - Home</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header><h1>FBS E-Books</h1><p>World's Best
Online Books Website</p></header>
<nav>
  <a href="home.html">Home</a> <a
href="registration.html">Registration</a> <a
href="login.html">Login</a> <a
href="catalog.html">Catalog</a> <a
href="cart.html">Cart</a>
</nav>
<main>
  <section class="welcome">
    <h2>Welcome to FBS E-Books</h2>
    <p>Find your favourite books at affordable
prices!</p>
    <a class="button" href="catalog.html">Explore
Books</a>
  </section>
  <h2 class="center">Why Shop With Us?</h2>
  <section class="features">
    <div class="box"><h3>Best Books</h3><p>A
wide collection of books.</p></div>
    <div class="box"><h3>Best
Prices</h3><p>Books at affordable
prices.</p></div>
    <div class="box"><h3>Easy
Shopping</h3><p>Simple and quick
shopping.</p></div>
  </section>
</main>
<footer><p>© 2026 FBS E-Books. All Rights
Reserved.</p></footer>
</body>
</html>
```

login.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-
width, initial-scale=1.0">
<title>Login - FBS</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header><h1>FBS E-Books</h1><p>World's Best
Online Books Website</p></header>
<nav>
  <a href="home.html">Home</a> <a
href="registration.html">Registration</a> <a
href="login.html">Login</a> <a
href="catalog.html">Catalog</a> <a
href="cart.html">Cart</a>
</nav>
<main class="form-box">
  <h2 class="center">User Login</h2>
  <form>
    <label>Username:</label>
    <input type="text" name="username"
placeholder="Enter username" required>
    <label>Password:</label>
    <input type="password" name="password"
placeholder="Enter password" required>
    <p><input class="button" type="submit"
value="Login"> <input class="reset-button"
type="reset" value="Reset"></p>
  </form>
  <p class="center">New user? <a
href="registration.html">Register here</a></p>
</main>
<footer><p>© 2026 FBS E-Books. All Rights
Reserved.</p></footer>
</body>
</html>
```

Registran.html

```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-
width, initial-scale=1.0">
<title>Registration - FBS</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header><h1>FBS E-Books</h1><p>World's Best
Online Books Website</p></header>
<nav>
  <a href="home.html">Home</a> <a
href="registration.html">Registration</a> <a
href="login.html">Login</a> <a
href="catalog.html">Catalog</a> <a
href="cart.html">Cart</a>
</nav>
<main class="form-box">
  <h2 class="center">Registration Form</h2>
  <form>
    <label>Name:</label>
    <input type="text" name="username"
placeholder="Enter your name" required>
    <label>Password:</label>
    <input type="password" name="password"
placeholder="Enter password" required>
    <label>Email ID:</label>
    <input type="email" name="email"
placeholder="Enter email" required>
    <label>Gender:</label>
    <input type="radio" name="gender"
value="Male"> Male
    <input type="radio" name="gender"
value="Female"> Female
    <label>Languages Known:</label>
    <input type="checkbox" name="language"
value="English"> English
    <input type="checkbox" name="language"
value="Telugu"> Telugu
    <input type="checkbox" name="language"
value="Hindi"> Hindi
    <p><input class="button" type="submit"
value="Register"> <input class="reset-button"
type="reset" value="Reset"></p>
  </form>
</main>
<footer><p>© 2026 FBS E-Books. All Rights
Reserved.</p></footer>
</body>
</html>
```

style.css

```
body {
  font-family: Arial;
  margin: 0;
  background: #f2f2f2;
}
header,
footer {
  background: #263b80;
  color: white;
  text-align: center;
  padding: 12px;
}
nav {
  background: #222;
  padding: 15px;
  text-align: center;
}
nav a {
  color: white;
  text-decoration: none;
  margin: 12px;
}

main {
  background: white;
  margin: 20px;
  padding: 25px;
  border-radius: 8px;
}

.center {
  text-align: center;
}

.welcome {
  text-align: center;
  padding: 25px;
}

.features {
  display: flex;
  gap: 15px;
  justify-content: center;
}

.book-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit,
minmax(180px, 1fr));
  gap: 15px;
}

.box {
```

```
  background: #eef2ff;
  padding: 18px;
  text-align: center;
  border: 1px solid #ccc;
  border-radius: 6px;
}

.form-box {
  max-width: 450px;
  margin: 25px auto;
}

label {
  display: block;
  margin-top: 12px;
  font-weight: bold;
}

input[type="text"],
input[type="password"],
input[type="email"],
textarea {
  width: 100%;
  padding: 9px;
  margin-top: 5px;
  box-sizing: border-box;
}

.button,
.reset-button {
  display: inline-block;
  background: #378026;
  color: white;
  padding: 10px 15px;
  border: 0;
  text-decoration: none;
  border-radius: 4px;
  cursor: pointer;
}
.reset-button {
  background: #666;
}
table {
  width: 100%;
  border-collapse: collapse;
  text-align: center;
}
th,
td {
  border: 1px solid #888;
  padding: 12px;
}
th {
  background: #dce5ff;
}
```

Week – 2

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-
width, initial-scale=1">
  <title>FBS E-Books</title>
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3
/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body class="bg-light">

  <header class="text-center text-white p-3"
style="background:#263b80">
    <h1>FBS E-Books</h1>
    <p class="mb-0">World's Best Online Books
Website</p>
  </header>

  <ul class="nav nav-pills justify-content-center bg-
dark p-2">
    <li class="nav-item"><button class="nav-link
active" data-bs-toggle="pill" data-bs-
target="#home">Home</button></li>
    <li class="nav-item"><button class="nav-link"
data-bs-toggle="pill" data-bs-
target="#register">Registration</button></li>
    <li class="nav-item"><button class="nav-link"
data-bs-toggle="pill" data-bs-
target="#login">Login</button></li>
    <li class="nav-item"><button class="nav-link"
data-bs-toggle="pill" data-bs-
target="#catalog">Catalog</button></li>
    <li class="nav-item"><button class="nav-link"
data-bs-toggle="pill" data-bs-
target="#cart">Cart</button></li>
  </ul>

  <main class="container bg-white rounded
shadow-sm my-4 p-4 tab-content">

    <section class="tab-pane fade show active text-
center" id="home">
      <h2>Welcome to FBS E-Books</h2>
      <p>Shopping at FBS can be both fun and
economical.<br>Find your favourite books at
affordable prices!</p>
      <button class="btn btn-success mb-4" data-bs-
toggle="pill" data-bs-target="#catalog">Explore
Books</button>
      <h3>Why Shop With Us?</h3>
      <div class="row g-3">
```

```
      <div class="col-md-4"><div class="card card-
body"><h5>Best Books</h5>A wide collection of
books.</div></div>
      <div class="col-md-4"><div class="card card-
body"><h5>Best Prices</h5>Books at affordable
prices.</div></div>
      <div class="col-md-4"><div class="card card-
body"><h5>Easy Shopping</h5>Simple and quick
shopping.</div></div>
    </div>
  </section>

  <section class="tab-pane fade mx-auto"
id="register" style="max-width:450px">
    <h2 class="text-center">Registration
Form</h2>
    <form>
      <input class="form-control mb-2" type="text"
placeholder="Name" required>
      <input class="form-control mb-2"
type="password" placeholder="Password"
required>
      <input class="form-control mb-2"
type="email" placeholder="Email ID" required>
      <div class="mb-2">Gender:
        <label class="ms-2"><input type="radio"
name="g"> Male</label>
        <label class="ms-2"><input type="radio"
name="g"> Female</label>
      </div>
      <button class="btn btn-
success">Register</button>
      <button type="reset" class="btn btn-
secondary">Reset</button>
    </form>
  </section>

  <section class="tab-pane fade mx-auto"
id="login" style="max-width:450px">
    <h2 class="text-center">User Login</h2>
    <form>
      <input class="form-control mb-2" type="text"
placeholder="Username" required>
      <input class="form-control mb-3"
type="password" placeholder="Password"
required>
      <button class="btn btn-
success">Login</button>
      <button type="reset" class="btn btn-
secondary">Reset</button>
    </form>
    <p class="text-center mt-3">New user? <a
href="#" data-bs-toggle="pill" data-bs-
target="#register">Register here</a></p>
  </section>
```

```
   <section class="tab-pane fade text-center"
id="catalog">
     <h2>Book Catalog</h2>
     <div class="row g-3">
       <div class="col-md-4"><div class="card card-
body"><h5>Java Programming</h5><p>Price:
₹500</p><button class="btn btn-success" data-bs-
toggle="pill" data-bs-target="#cart">Add to
Cart</button></div></div>
       <div class="col-md-4"><div class="card card-
body"><h5>Python Programming</h5><p>Price:
₹400</p><button class="btn btn-success" data-bs-
toggle="pill" data-bs-target="#cart">Add to
Cart</button></div></div>
       <div class="col-md-4"><div class="card card-
body"><h5>Web Technologies</h5><p>Price:
₹300</p><button class="btn btn-success" data-bs-
toggle="pill" data-bs-target="#cart">Add to
Cart</button></div></div>
     </div>
   </section>

   <section class="tab-pane fade text-center"
id="cart">
     <h2>Shopping Cart</h2>
     <table class="table table-bordered">
       <tr class="table-primary"><th>Book
Name</th><th>Price</th><th>Quantity</th><th>
Amount</th></tr>
       <tr><td>Java
Programming</td><td>₹500</td><td>1</td><td>₹
500</td></tr>
       <tr><td>Python
Programming</td><td>₹400</td><td>2</td><td>₹
800</td></tr>
       <tr class="table-primary"><th
colspan="3">Total
Amount</th><th>₹1300</th></tr>
     </table>
     <button class="btn btn-success" data-bs-
toggle="pill" data-bs-target="#catalog">Continue
Shopping</button>
   </section>

  </main>
  <footer class="text-center text-white p-3"
style="background:#263b80">© 2026 FBS E-
Books. All Rights Reserved.</footer>

  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/
dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Week – 3

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration - FBS</title>
</head>
<body>
  <h2>Registration Form</h2>
  <form onsubmit="return validate()">
    Name:<br>
    <input type="text" id="name"><br><br>
    Password:<br>
    <input type="password"
id="password"><br><br>
    Email:<br>
    <input type="text" id="email"><br><br>
    Gender:<br>
    <input type="radio" name="gender"> Male
    <input type="radio" name="gender"> Female
    <br><br>
    <input type="submit" value="Register">
  </form>
  <script>
    function validate() {
      var name =
document.getElementById("name").value;
      var password =
document.getElementById("password").value;
      var email =
document.getElementById("email").value;
      if (name == "") {
        alert("Please enter your name");
        return false;
      }
      if (password.length < 6) {
        alert("Password must contain at least 6
characters");
        return false;
      }
      if (!email.includes("@")) {
        alert("Please enter a valid email");
        return false;
      }
      alert("Registration Successful");
      return true;
    }
  </script>
</body>
</html>
```

Week – 4

```
<!DOCTYPE html>
<html>
<head>
 <title>Weather Application</title>
 <script
src="https://cdn.jsdelivr.net/npm/chart.js"></scrip
t>
</head>
<body style="font-family:Arial; text-
align:center;">
 <h1>Weather Information</h1>
 <input type="text" id="city" placeholder="Enter
city name">
 <button onclick="getWeather()">Get
Weather</button>
 <h2 id="result"></h2>
 <p id="details"></p>
 <div style="width:600px; margin:auto;">
  <canvas id="weatherChart"></canvas>
 </div>
 <script>
  const apiKey =
"4f5d07bea1eeb6e322d44e1d1e7b062c";

  async function getWeather() {
   const city =
document.getElementById("city").value;

   if (city === "") {
    alert("Please enter city name");
    return;
   }

   try {
    const url =
`https://api.openweathermap.org/data/2.5/forecast?
q=${city}&appid=${apiKey}&units=metric`;
    const response = await fetch(url);
    const data = await response.json();

    if (!response.ok) {
     throw new Error(data.message || "Unable to
get weather");
    }

    document.getElementById("result").innerHT
ML = "Weather in " + city;
    document.getElementById("details").innerHT
ML =
     "Temperature: " + data.list[0].main.temp + "
°C<br>" +
     "Humidity: " + data.list[0].main.humidity +
"%";
```

```
      drawGraph(data.list);
    } catch(error) {
      alert(error.message);
    }
  }

  function drawGraph(list) {
    const labels = [];
    const temperatures = [];

    for (let i = 0; i < 8; i++) {
      labels.push(list[i].dt_txt);
      temperatures.push(list[i].main.temp);
    }

    const ctx =
document.getElementById("weatherChart");

    if (window.myChart) {
      window.myChart.destroy();
    }

    window.myChart = new Chart(ctx, {
      type: "line",
      data: {
        labels: labels,
        datasets: [{
          label: "Temperature (°C)",
          data: temperatures,
          borderWidth: 2
        }]
      }
    });
  }
 </script>
</body>
</html>
```

Week – 5

bookstore.dtd

```
<!ELEMENT bookstore (book+)>
<!ELEMENT book (title, author, price, year)>
<!ELEMENT title (#PCDATA)>
<!ELEMENT author (#PCDATA)>
<!ELEMENT price (#PCDATA)>
<!ELEMENT year (#PCDATA)>
```

bookstore.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<!-- xmllint -noout -valid bookstore.xml-->
<!DOCTYPE bookstore SYSTEM
"bookstore.dtd">

<!-- xmllint -noout -schema bookstore.xsd
bookstore.xml-->
<bookstore>
 <book>
  <title>Java Programming</title>
  <author>James Gosling</author>
  <price>450</price>
  <year>2024</year>
 </book>
 <book>
  <title>Python Basics</title>
  <author>Guido Rossum</author>
  <price>350</price>
  <year>2023</year>
 </book>
 <book>
  <title>Web Development</title>
  <author>John Smith</author>
  <price>500</price>
  <year>2025</year>
 </book>
</bookstore>
```

bookstore.xsd

```
<?xml version="1.0" encoding="UTF-8"?>

<xs:schema
xmlns:xs="http://www.w3.org/2001/XMLSchema"
>
 <xs:element name="bookstore">
  <xs:complexType>
   <xs:sequence>
    <xs:element name="book"
```

```
maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="title"
type="xs:string"/>
            <xs:element name="author"
type="xs:string"/>
            <xs:element name="price"
type="xs:decimal"/>
            <xs:element name="year"
type="xs:integer"/>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
 </xs:element>
</xs:schema>
```

Week – 6

```
const express = require('express');
const cookieParser = require('cookie-parser');
const session = require('express-session');

const app = express();

app.use(cookieParser());

app.use(session({
   secret: 'abc',
   resave: false,
   saveUninitialized: true
}));

//npm init -y
//npm install express cookie-parser express-session
// Home
app.get('/', (req, res) => {
   res.send('<h1>Session Tracking</h1><a
href="/cookie">Cookie</a><br><a
href="/session">Session</a>');
});

// Create Cookie
app.get('/cookie', (req, res) => {
   res.cookie('name', 'Sunny');
   res.send('Cookie Created');
});

// Create Session
app.get('/session', (req, res) => {
   req.session.name = 'Sunny';
   res.send('Session Created: ' + req.session.name);
});

// Destroy Session
app.get('/logout', (req, res) => {
   req.session.destroy();
   res.send('Session Destroyed');
});

app.listen(3000, () => {
   console.log('Server running on port 3000');
});
```

Week – 7

week7

```
// Importing required modules
const http = require('http');
const os = require('os');
const path = require('path');
const EventEmitter = require('events');

// Creating EventEmitter object
const event = new EventEmitter();

// Registering an event
event.on('start', () => {


   console.log('Server Started');
});

// Displaying operating system information
console.log("OS:", os.platform());

// Displaying file name
console.log("File:", path.basename(__filename));

// Creating HTTP server
const server = http.createServer((req, res) => {
   res.end("Welcome to Node.js Server");
});

// Starting server
server.listen(7000, () => {
   event.emit('start');
   console.log("Server running at
http://localhost:7000");
});
```

week8

```
const express = require('express');
const app = express();

app.use(express.json());

let students = [
   { id: 1, name: "Rahul", age: 20 },
   { id: 2, name: "Priya", age: 21 }
];

// CREATE
app.post('/students', (req, res) => {
   students.push(req.body);
   res.send("Student Added");
});

// READ
app.get('/students', (req, res) => {
   res.json(students);
});

// UPDATE
app.put('/students/:id', (req, res) => {

   let id = Number(req.params.id);

   let student = students.find(s => s.id === id);

   if (student) {
      student.name = req.body.name;
      student.age = req.body.age;

      res.send("Student Updated");
   } else {
      res.send("Student Not Found");
   }
});

// DELETE
app.delete('/students/:id', (req, res) => {
   students = students.filter(s => s.id !=
req.params.id);
   res.send("Student Deleted");
});

app.listen(7000, () => {
   console.log("Server running at
http://localhost:7000");
});
```

week9

```
const express = require('express');
const jwt = require('jsonwebtoken');

const app = express();
app.use(express.json());

const SECRET = "mysecretkey";

// LOGIN
app.post('/login', (req, res) => {
   const token = jwt.sign({ user: "admin" },
SECRET);
   res.json({ token });
});

// JWT AUTHENTICATION MIDDLEWARE
function verifyToken(req, res, next) {
   const token = req.headers.authorization;

   if (!token) {
      return res.sendStatus(401);
   }

   try {
      jwt.verify(token.split(' ')[1], SECRET);
      next();
   } catch {
      res.sendStatus(403);
   }
}

// STUDENT DATA
let students = [
   { id: 1, name: "Rahul", age: 20 },
   { id: 2, name: "Priya", age: 21 }
];

// CREATE
app.post('/students', verifyToken, (req, res) => {
   students.push(req.body);
   res.send("Student Added");
});

// READ
app.get('/students', verifyToken, (req, res) => {
   res.json(students);
});

// UPDATE
app.put('/students/:id', verifyToken, (req, res) => {
   const id = Number(req.params.id);

   const student = students.find(s => s.id === id);
```

```
    if (student) {
        student.name = req.body.name;
        student.age = req.body.age;
        res.send("Student Updated");
    } else {
        res.status(404).send("Student Not Found");
    }
});

// DELETE
app.delete('/students/:id', verifyToken, (req, res) =>
{
    const id = Number(req.params.id);

    students = students.filter(s => s.id !== id);

    res.send("Student Deleted");
});

// START SERVER
app.listen(7000, () => {
    console.log("Server running at
http://localhost:7000");
});
```

week10
app.jsx

```
import React, { useState } from "react";
import './app.css'
function App() {

  const [task, setTask] = useState("");
  const [tasks, setTasks] = useState([]);

  function addTask() {
    if (task.trim() !== "") {
      setTasks([...tasks, task]);
      setTask("");
    }
  }

  function deleteTask(index) {
    setTasks(tasks.filter((_, i) => i !== index));
  }

  return (
    <div>
      <h1>TODO List</h1>

      <input
        type="text"
        placeholder="Enter task"
        value={task}
        onChange={(e) => setTask(e.target.value)}
      />

      <button onClick={addTask}>Add
Task</button>

      <h3>My Tasks</h3>

      {tasks.map((t, index) => (
        <p key={index}>
          {t}
          <button onClick={() =>
deleteTask(index)}>
            Delete
          </button>
        </p>
      ))}
    </div>
  );
}

export default App;
```

app.css

```
div {
  text-align: center;
  margin: 30px;
}

input {
  padding: 10px;
}

button {
  padding: 10px;
  margin: 5px;
  background-color: blue;
  color: white;
  border: none;
  cursor: pointer;
}
```

main.jsx

```
import React from 'react'
import ReactDOM from 'react-dom/client'
import App from './app.jsx'
import './app.css'

ReactDOM.createRoot(document.getElementById(
'root')).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>,
)
```